CPE 381 Classwork 10: Fourier Series

Student Name (as in Canvas): _____

A Number: _____

Points: 20 Duration: 15 minutes Due: September 30, 2026 (In-Class)

1. Find the Fourier coefficients for $x(t) = B + A\sin(\Omega_0 t + \theta)$ and hence write the fourier representation in terms of complex exponentials. **(10 Points)**

Answer

$$\sin(\Omega_0 t+\theta) = \frac{1}{2j}\left[e^{j(\Omega_0 t+\theta)} - e^{-j(\Omega_0 t+\theta)}\right]$$

$$= -\frac{1}{2j}e^{-j\Omega_0 t}e^{-j\theta} + \frac{1}{2j}e^{j\Omega_0 t}e^{j\theta}$$

Hence $x(t) = B - \frac{A}{2j}e^{-j\Omega_0 t}e^{-j\theta} + \frac{A}{2j}e^{j\Omega_0 t}e^{j\theta}$

is the Fourier series representation.

Comparing it with $x(t) = \sum_{k=-\infty}^{\infty} X_k e^{jk\Omega_0 t}$

$X_0 = B$

$X_{-1} = -\frac{A}{2j}e^{-j\theta}$

$X_1 = \frac{A}{2j}e^{j\theta}$

$X_k = 0$ for $k \neq \{-1, 0, 1\}$

2. $x(t) = \cos(2t + \frac{\pi}{4})$. Find the Fourier series representation by finding the Fourier series coefficients. In order to do that, first, write down the value of fundamental frequency Ω_0. **(8 + 2 Points)**

Answer

$\Omega_0 = 2$, $T_0 = \frac{2\pi}{2} = \pi$

$x(t) = \cos\left(2t + \frac{\pi}{4}\right) = \frac{1}{2}\left[e^{j(2t+\frac{\pi}{4})} + e^{-j(2t+\frac{\pi}{4})} \right]$

($\Omega_0 \to k$)

$= \frac{1}{2} e^{j(2)\cdot 1 \cdot t} e^{j\pi/4} + \frac{1}{2} e^{j2\cdot(-1)\cdot t} e^{-j\pi/4}$

Hence X_k for $k=1$ i.e. $X_1 = \frac{1}{2} e^{j\pi/4}$

$= \frac{1}{2}\left[\frac{1+j}{\sqrt{2}}\right]$

X_k for $k=-1$, i.e. $X_{-1} = \frac{1}{2} e^{-j\pi/4}$

$= \frac{1}{2}\left[\frac{1-j}{\sqrt{2}}\right]$

$X_k = 0$ for $k \neq \{-1, 1\}$